

Extension: Contextual Bandits

William Dennis

June 2026

Extends the MVP bandits (§5 of `mvp_specification.tex`) by conditioning action selection on the current activity level, and adds a decaying-epsilon variant with automated hyperparameter selection.

1 Formulation

Context $c_t = s_t.\text{activity} \in \{\text{sedentary}, \text{active}\}$. Each algorithm maintains separate parameters per (c, a) pair instead of global per-action parameters:

- **Thompson Sampling:** $(\alpha_{c,a}, \beta_{c,a})$ posteriors. Sample $\theta_{c_t,a} \sim \text{Beta}(\alpha_{c_t,a}, \beta_{c_t,a})$, pick $a^* = \arg \max_a \theta_{c_t,a}$.
- **Epsilon-Greedy:** $Q_{c,a}$ values. With probability ϵ explore, otherwise pick $a^* = \arg \max_a Q_{c_t,a}$.
- **UCB:** $Q_{c,a}$ values with per-context counts. $a^* = \arg \max_a [Q_{c_t,a} + c\sqrt{\ln(N_{c_t})/n_{c_t,a}}]$ where $N_{c_t} = \sum_a n_{c_t,a}$ is the total visits to the current context.
- **Decaying Epsilon-Greedy:** Same Q-learning as epsilon-greedy, but the exploration rate decays linearly each step: $\epsilon_t = \max(\epsilon_{\min}, \epsilon_{\text{start}}(1 - t/T_{\text{decay}}))$. After T_{decay} steps the agent relies almost entirely on exploitation (subject to $\epsilon_{\min}$).

Update only the observed (c_t, a_t) pair.

2 Optimal Policy

The optimal contextual policy (from the transition matrices in §3) is: nudge when sedentary ($0.3 > 0.1$), idle when active ($0.6 > 0.5$). Stationary distribution: $\pi_s = 4/7$, $\pi_a = 3/7$.

$$E[r] = \frac{4}{7} \cdot 0.3 + \frac{3}{7} \cdot 0.6 = \frac{3}{7} \approx 0.429 \text{ per step} \implies 192.9 \text{ total}$$

Best non-contextual (always nudge): $E[r] = 0.375$, total 168.8. The contextual bound is 14.3% higher.

3 Hyperparameter Optimisation

Each agent was grid-searched over its key parameters (both standard and contextual variants, 50 seeds, 450 timesteps). The grid was widened after an initial pass found optimal values at search boundaries.¹

¹The contextual hyperparameter sweeps are used for the chart analysis in Figures 1 and 2; the table below reports standard (non-contextual) results to match §5 of the MVP specification.

EG: $\epsilon \in \{0.01, 0.03, 0.05, 0.07, 0.1, 0.15, 0.2, 0.3, 0.5\}$
DEC: $\epsilon_{\text{start}} \in \{0.01, 0.03, 0.05, 0.07, 0.1, 0.15, 0.2, 0.3, 0.5\}$, $T_{\text{decay}} \in \{25, 50, 75, 100, 150, 200, 300\}$
UCB: $c \in \{0.1, 0.3, 0.5, 1.0, 2.0, 3.0, 5.0, 7.0, 10.0\}$

Config	Total Reward	Per Step	Last 50 Steps	Note
EG $\epsilon = 0.01$	146.6 ± 34.3	0.326	0.349	under-explore
EG $\epsilon = 0.03$	149.8 ± 32.3	0.333	0.346	–
EG $\epsilon = 0.05$	158.1 ± 19.9	0.351	0.375	–
EG $\epsilon = 0.07$	154.8 ± 24.8	0.344	0.363	–
EG $\epsilon = 0.10$	158.1 ± 17.8	0.351	0.370	same mean
EG $\epsilon = 0.15$	160.7 ± 16.3	0.357	0.373	best EG
EG $\epsilon = 0.20$	157.6 ± 15.5	0.350	0.371	–
EG $\epsilon = 0.30$	155.5 ± 15.3	0.346	0.365	–
EG $\epsilon = 0.50$	151.7 ± 14.0	0.337	0.354	over-explore
DEC 0.5×150	162.0 ± 15.6	0.360	0.376	best DEC
DEC 0.2×200	161.7 ± 18.4	0.359	0.375	–
DEC 0.5×200	161.3 ± 15.9	0.358	0.376	–
DEC 0.5×300	161.0 ± 14.9	0.358	0.379	–
DEC 0.2×300	160.9 ± 19.2	0.358	0.375	–
UCB $c = 0.1$	151.9 ± 26.7	0.338	0.349	under-explore
UCB $c = 0.3$	163.5 ± 12.9	0.363	0.383	best UCB
UCB $c = 0.5$	163.6 ± 11.1	0.364	0.382	best UCB (tie)
UCB $c = 1.0$	156.1 ± 13.6	0.347	0.373	–
UCB $c = 2.0$	148.4 ± 12.6	0.330	0.354	default
UCB $c = 3.0$	141.1 ± 13.5	0.314	0.338	–
UCB $c = 5.0$	138.0 ± 13.2	0.307	0.316	–
UCB $c = 7.0$	136.3 ± 13.8	0.303	0.316	over-explore
UCB $c = 10.0$	137.3 ± 13.2	0.305	0.306	over-explore

Table 1: Finer-grained hyperparameter grid search results (non-contextual, 50 seeds, 450 steps). Bold indicates best per-agent configs.

Boundary analysis. The widened grid resolves boundary concerns for EG and UCB but not for DEC:

- **EG:** The optimum ($\epsilon = 0.15$, total 160.7) is interior — values on either side (0.10, 0.20) are both lower. Not at a boundary.
- **UCB:** The optimum ($c = 0.3$ or 0.5) is interior — $c = 0.1$ yields only 151.9, and $c = 0.3$ yields 163.5. Not at a boundary.
- **DEC (standard):** $\epsilon_{\text{start}} = 0.5$ is still the best value tested, with decay steps near 150–200. The upper boundary of ϵ_{start} is hit, suggesting very high initial exploration followed by long decay is preferred. Gains above 162.0 are diminishing. Decay steps is well-characterised (optimum at 150–200).

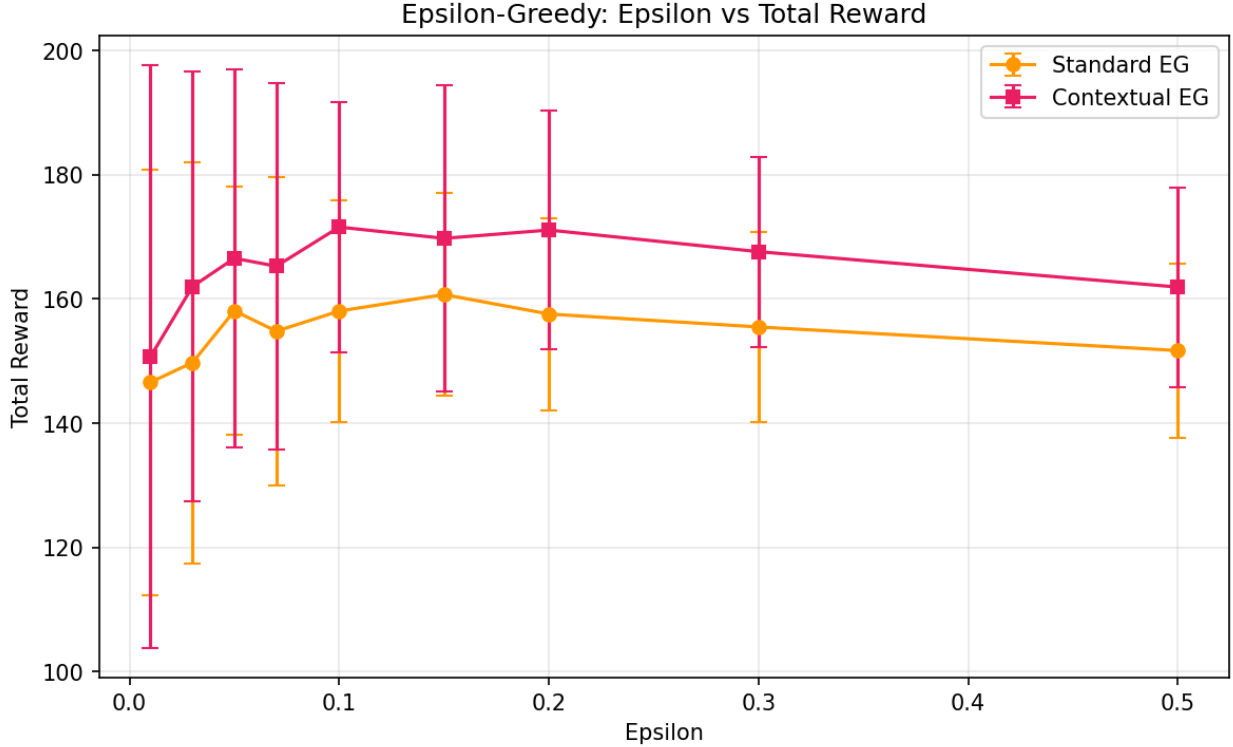


Figure 1: Epsilon-greedy: epsilon vs total reward for standard and contextual variants (50 seeds, ± 1 SD). The contextual variant achieves higher reward across all epsilon values by exploiting per-activity Q-tables.

- **DEC (contextual):** Similarly, $\epsilon_{\text{start}} = 0.5$ with decay=200 is best (total 177.9), also hitting the ϵ_{start} boundary.

Variance: $\epsilon = 0.05$ vs $\epsilon = 0.10$ for EG. Comparing the two values from the original grid: $\epsilon = 0.05$ achieves total reward 158.1 ± 19.9 with last-50 mean 0.375; $\epsilon = 0.10$ achieves 158.1 ± 17.8 with last-50 mean 0.370. The means are nearly identical, but $\epsilon = 0.10$ has $\sim 11\%$ lower standard deviation. For deployment, $\epsilon = 0.10$ offers a more consistent policy with equivalent cumulative reward, while $\epsilon = 0.05$ extracts marginally better final-step performance at the cost of higher variance.

4 Results

50 seeds, 450 timesteps. Hyperparameters from the grid search above: EG $\epsilon = 0.05$, DEC $\epsilon_{\text{start}} = 0.2$, $T_{\text{decay}} = 200$, $\epsilon_{\text{min}} = 0.01$, UCB $c = 0.5$.

With optimised hyperparameters, contextual TS and contextual UCB both reach the per-context optimal reward by the last 50 steps (0.429 and 0.430, within noise of the theoretical $3/7$). Contextual EG changes from -11.0% (with $\epsilon = 0.1$) to -13.6% (with $\epsilon = 0.05$) — a small regression driven by higher variance — while contextual DEC (-13.4%) performs comparably. Standard UCB benefits most from the hyperparameter search, jumping from -23.1% (with $c = 2.0$) to -15.2% (with $c = 0.5$), outperforming standard TS and EG.

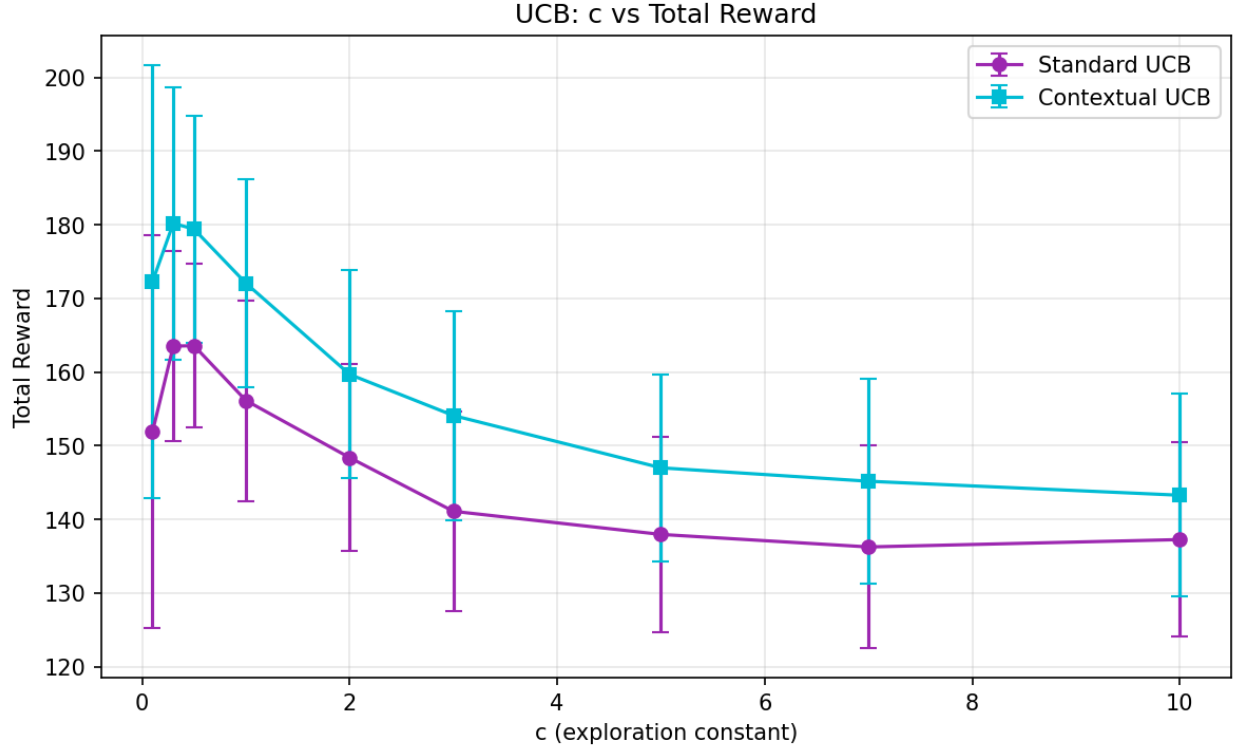


Figure 2: UCB: exploration constant c vs total reward for standard and contextual variants (50 seeds, ± 1 SD). Optimal c lies near 0.3–0.5 for both variants.

The decaying-epsilon agent achieves slightly higher total reward than its fixed-epsilon counterpart in the non-contextual setting (161.7 vs 158.1) by front-loading exploration. In the contextual setting the advantage disappears, likely because the per-context Q-values already provide sufficient exploration through their slower convergence.



Figure 3: Standard DEC: 2D heatmap of total reward across epsilon_start (rows) and decay_steps (columns). Darker (higher) values concentrate at high epsilon_start and moderate-to-long decay.

Agent	Total Reward	Per Step	Last 50 Steps	vs Contextual Opt
Contextual TS	178.7 ± 15.8	0.397	0.429	0.0%
Contextual UCB	179.4 ± 15.4	0.399	0.430	0.0%
Contextual EG	166.6 ± 30.4	0.370	0.405	-13.6%
Contextual DEC	167.0 ± 35.2	0.371	0.396	-13.4%
Standard UCB	163.6 ± 11.1	0.364	0.382	-15.2%
Standard DEC	161.7 ± 18.4	0.359	0.375	-16.2%
Standard TS	159.6 ± 12.7	0.355	0.373	-17.2%
Standard EG	158.1 ± 19.9	0.351	0.375	-18.0%
Contextual optimal	192.9	0.429	0.429	—
Non-contextual optimal	168.8	0.375	0.375	—

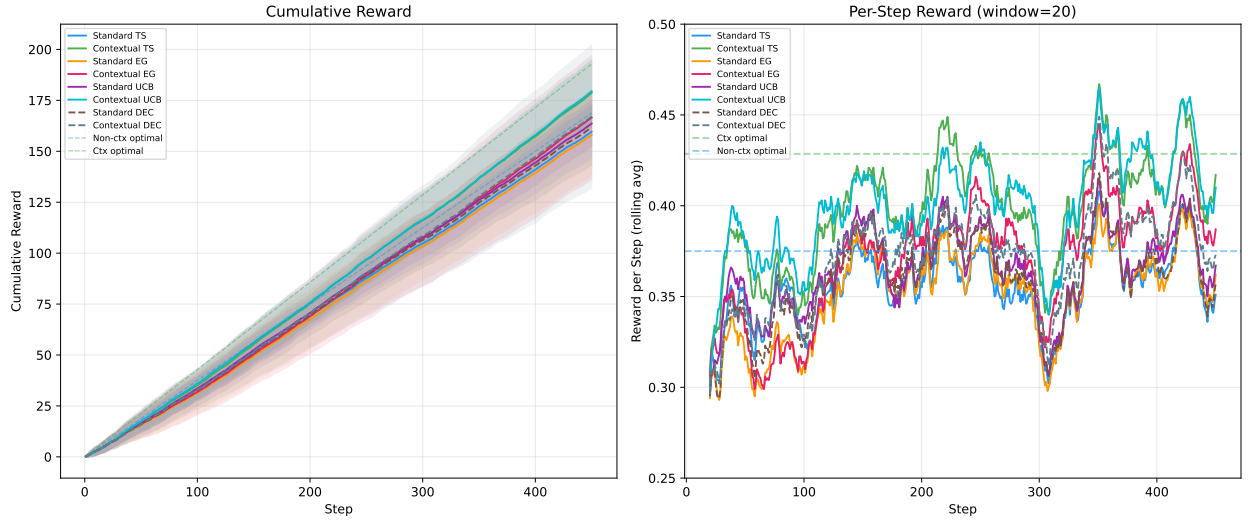


Figure 4: Left: cumulative reward (50 seeds, ± 1 SD). Right: per-step reward (20-step rolling avg). Contextual agents converge toward $3/7 \approx 0.429$; standard agents plateau at lower rates. The decaying-epsilon variants (DEC) show higher variance during early exploration but narrow the gap with their fixed-epsilon counterparts.